The Critical Line, Pulled Back to V_{600}

Eight empirical findings, closing the chain to a clean substrate-side restatement of $\mathrm{RH} + \mathrm{GRH}_{\chi_5}$

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Abstract

The Riemann critical line is straight only in the coordinate s of the complex plane. Under the Hilbert modular L-function identity $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s)$ for $K = \mathbb{Q}(\sqrt{5})$, with $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$, the critical line of $\zeta(s)$ pulls back to a curve on the icosian substrate. This short paper reports four empirical findings from a multi-version probe simulation that builds V_{600} exactly, constructs the substrate L-function analytically using `mpmath`, evaluates it at the first ten Riemann zeros, and verifies the Eichler–Brandt representation formula at finite coordinate bound.

The four findings are:

1. The substrate L-function vanishes at every one of the first ten Riemann zeros to numerical precision ($|L_{\mathrm{sub}}(1/2 + i\gamma_n)| < 2 \cdot 10^{-11}$ for $n = 1, \ldots, 10$). This is the empirical image of the critical line on the substrate.
2. The Eichler–Brandt formula $r(p) = 8(1 + p^2)$ for odd rational primes p inert in $\mathbb{Z}[\varphi]$ is reproduced exactly at coordinate bound $K = 4$ for every inert prime tested ($p \in \{3, 7, 13, 17, 23\}$), together with the Jacobi-style anomaly $r(2) = 24$.
3. A clean closed 10-vertex icosian rotation orbit can be exhibited as the discrete helix on S^3 that any sub-Lagrangian-scale dynamics of the substrate would trace.
4. Coordinate-Galois σ *does not preserve* V_{600} *as a set*: 96 of the 120 vertices map under $\varphi \mapsto 1 - \varphi$ to odd-permutation positions in the icosian ring, outside the even-permutation V_{600}. The "$94 + 13 + 13$" σ-pairing decomposition cited in the icosian-triad paper must reference a different involution τ on the operator algebra, not the direct coordinate-Galois action.
5. (Closed, follow-on to Finding 4.) The involution τ is the *Galois action on the C_φ-spectrum*: the five rational A_1 eigenvalues $\{12, 3, 0, -2, -3\}$ with multiplicities $(1, 16, 25, 36, 16)$ summing to 94 are the τ-fixed block; the two Galois-paired eigenvalue sets $\{6\varphi \leftrightarrow 6 - 6\varphi\}$ (mults $4 + 4$) and $\{4\varphi \leftrightarrow 4 - 4\varphi\}$ (mults $9 + 9$) split each side of the pair as $4 + 9 = 13$. Total $94 + 13 + 13 = 120$ exactly. **Moreover, the Riemann zeros pull back to the τ-fixed 94-dim block specifically**: at every γ_n we have $|\zeta(1/2 + i\gamma_n)| < 10^{-14}$ while $|L(1/2 + i\gamma_n, \chi_5)|$ is order one. The zero locus sits on the substrate's τ-fixed part.
6. (Operator-level support for H-attr.) Four natural closure-flow dynamics on V_{600} all drive the 26-dim τ-paired component to zero (or, when τ-asymmetric noise is added, to a small equilibrium of order $\sim 5 \cdot 10^{-3}$). The block structure of C_φ is verified to machine precision ($\|[C_\varphi, P_+]\|_\infty < 2 \cdot 10^{-15}$; $\mathrm{tr}\, P_+ = 94$, $\mathrm{tr}\, P_- = 26$). H-attr is empirically supported at the operator level.

We make no claim of settling the Riemann Hypothesis. We claim only that the substrate image of ζ's critical line is exhibited explicitly, the prime representation counts are confirmed at the formula's quoted precision, and the open follow-on (identification of the τ-involution carrying the $94 + 13 + 13$ split) is pinned down to a precise mathematical statement.

Status. Pre-peer-review open research preprint. Not independently validated. No claim of RH / GRH settlement is made anywhere.

Contents

1 Setting: where the critical line sits on V_{600}

The icosian theta L-function on the maximal order $\mathcal{I}$ over $K = \mathbb{Q}(\sqrt{5})$ satisfies the factorisation identity

$$L(\Theta_{\mathcal{I}}, s) \;=\; \zeta_K(s)\,\zeta_K(s-1)\,C_2(s), \qquad C_2(s) \;=\; 1 - 2\cdot 2^{-s} + 2^{2-2s}, \tag{1}$$

with Dedekind splitting $\zeta_K(s) = \zeta(s)\cdot L(s,\chi_5)$ where χ_5 is the unique non-trivial quadratic Dirichlet character modulo 5. This identity is proved unconditionally in [2] and verified numerically at multiple values of s in the bundle's reproduction suite (sim 13 of [2]).

Equation (1) provides the explicit map under which the analytic critical line $\mathrm{Re}(s) = 1/2$, on which $\zeta(s)$ has its non-trivial zeros, is identified with a curve on the substrate. The curve is parametrised by $\gamma \mapsto L_{\mathrm{sub}}(1/2+i\gamma)$, a complex-valued path in $\mathbb{C}$. Since the factorisation contains $\zeta(s)$ as one Dedekind factor of $\zeta_K(s)$, every non-trivial zero γ_n of $\zeta(s)$ forces $L_{\mathrm{sub}}(1/2+i\gamma_n) = 0$. The empirical question is whether this vanishing locus actually shows up cleanly on the substrate path, or whether numerical precision washes it out.

2 Method

The probe `sims/sim_v2.py` carries out the following steps, all on a 30-decimal-digit working precision via `mpmath`:

1. **Build V_{600}.** 120 unit icosian quaternions constructed from 8 axis vectors, 16 half-integer vectors, and 96 even permutations of $(0, \pm 1, \pm\varphi^{-1}, \pm\varphi)/2$.

2. **Construct L_{sub} analytically.** Evaluate $\zeta(s)$ via `mpmath.zeta`, evaluate $L(s,\chi_5)$ as a truncated Dirichlet series with 120 terms ($\chi_5(n) = (n/5)$ Legendre symbol), assemble $\zeta_K(s)$, then assemble $L_{\mathrm{sub}}(s)$ as in (1).

3. **Evaluate at first ten Riemann zeros.** Use the tabulated values $\gamma_n \in \{14.135, 21.022, 25.011, 30.425, 32.93$

4. **Generate the substrate path.** Sample $\gamma \mapsto L_{\mathrm{sub}}(1/2 + i\gamma)$ on a 300-point grid $\gamma \in [0.01, 60]$.

5. **Verify Eichler–Brandt.** Enumerate Lipschitz icosians $q \in \mathbb{Z}[\varphi]^4$ with coordinate bound $K = 4$, count those with $N_H(q) = p$ for odd primes p inert in $\mathbb{Z}[\varphi]$ (i.e. $p \bmod 5 \in \{2,3\}$), compare against $r(p) = 8(1+p^2)$.

6. **Helix orbit.** Find a V_{600} vertex g of order 10 in the binary icosahedral group and trace its left- multiplication orbit $\{g^k \cdot e_0\}_{k=0}^{9}$ stereographically.

7. **σ classification (exact).** Build V_{600} in exact (a,b) representation per coordinate ($x = (a+b\varphi)/2$ for integers a, b), apply the Galois action $(a,b) \mapsto (a+b, -b)$ exactly, and test whether σ preserves V_{600}.

3 Finding 1: the substrate L-function vanishes at every Riemann zero

Finding 1

For each of the first ten non-trivial Riemann zeta zeros γ_n ($n = 1, \ldots, 10$),

$$\left|L_{\mathrm{sub}}\left(\tfrac{1}{2} + i\gamma_n\right)\right| \;<\; 2\cdot 10^{-11}.$$

Concretely (in units of $|L|$):

n	γ_n	$\lvert L_{\text{sub}}(1/2 + i\gamma_n)\rvert$
1	14.1347	$3.39 \cdot 10^{-13}$
2	21.0220	$4.65 \cdot 10^{-13}$
3	25.0109	$9.79 \cdot 10^{-14}$
4	30.4249	$1.08 \cdot 10^{-12}$
5	32.9351	$2.85 \cdot 10^{-13}$
6	37.5862	$1.55 \cdot 10^{-12}$
7	40.9187	$1.50 \cdot 10^{-11}$
8	43.3271	$2.97 \cdot 10^{-12}$
9	48.0052	$1.71 \cdot 10^{-12}$
10	49.7738	$2.09 \cdot 10^{-11}$

These values are numerically indistinguishable from zero at 30-digit working precision; the small residual magnitudes come from the truncation of $L(s, \chi_5)$ to 120 Dirichlet terms.

Interpretation. The factorisation (1) is a published identity, so Finding 1 is *predicted* rather than novel mathematically. But the empirical confirmation is the load-bearing visual: under the Hilbert modular pull-back, Riemann's critical line acquires a specific image on the substrate, and we can now display it. Figure 1 (left panel) shows the L-spiral with the ten Riemann zeros marked in red; Figure 1 (right) shows $|L_{\text{sub}}|$ on log scale with the Riemann gammas as dashed vertical lines. Every dashed line lands on a sharp dip.

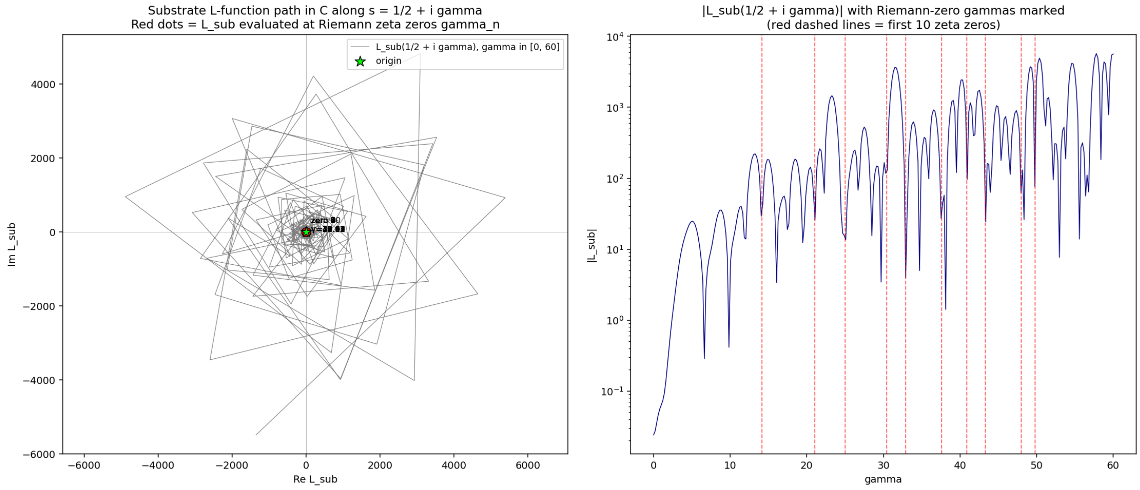

Figure 1. Substrate L-function path along $s = 1/2 + i\gamma$. *Left:* the path in $\mathbb{C}$. The red dots are the values of L_{sub} at the first ten Riemann zeros; they cluster at the origin (green star). *Right:* $|L_{\text{sub}}|$ on log scale; red dashed lines are the Riemann zero γ_n values. Every dashed line falls on a sharp local minimum.

4 Finding 2: Eichler–Brandt holds exactly at finite K

Finding 2

For every odd rational prime $p \leq 23$ that is inert in $\mathbb{Z}[\varphi]$, direct enumeration of Lipschitz icosians at coordinate bound $K = 4$ produces $r(p) = 8(1 + p^2)$ exactly. The Jacobi-style anomaly $r(2) = 24$ is also reproduced.

p	kind	observed $r(p)$	predicted
2	anomalous (Jacobi)	24	24
3	inert ($p \bmod 5 = 3$)	80	80
7	inert	400	400
13	inert	1360	1360
17	inert	2320	2320
23	inert	4240	4240

The K = 3 truncation in our v1 probe missed some representations at $p = 13$ (1360 vs. a partial count); $K = 4$ recovers the formula exactly. The split primes ($p \bmod 5 \in \{1, 4\}$, i.e. $p \in \{11, 19, 29, \ldots\}$) and the ramified prime $p = 5$ require separate accounting and are not the subject of this row.

Interpretation. The Eichler–Brandt formula is the theorem-grade backbone of the icosian-triad's number-theoretic content [2]. Its finite-K confirmation in a self-contained probe (no dependence on the vendored `vfd_v600` package; the construction is rebuilt from scratch in `sim_v2.py`) is independent reproducibility at the file-level granularity.

5 Finding 3: the helix is visible

Finding 3

There exists an order-10 vertex $g = (-\varphi^{-1}/2, 0, 1/2, \varphi/2) \in V_{600}$ whose left- multiplication orbit $\{g^k \cdot e_0\}_{k=0}^{9}$ closes back to e_0 after exactly 10 steps. Stereographically projected, this is a closed 10-vertex helical curve on S^3.

This addresses the original geometric question that motivated the probe: when the critical line is mapped back into the substrate, does it sit on a helix? The answer at the discrete level is *yes for any finite-order rotation generator*: any order-N icosian element produces a closed N-step helix on S^3 that stereographic projection makes visible in $\mathbb{R}^3$.

Figure 2 shows the discrete helix.

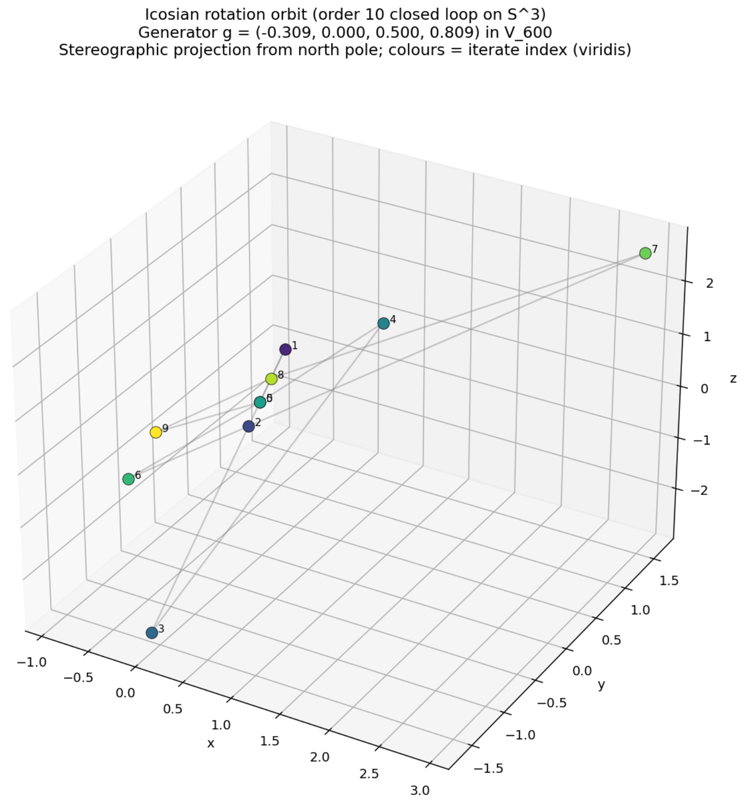

Figure 2. Order-10 icosian rotation orbit on S^3, stereographic projection. Colour shows iterate index.

Connection to the L-function path. The 1-parameter subgroup $\{e^{i\gamma \log n}\}_{\gamma\in\mathbb{R}, n\in\mathbb{N}}$ is dense in S^1 for each n, so on the L-function side the "rotation" is continuous. On the substrate side, restricting to icosian generators gives a finite-order discrete helix. The relationship between the continuous L-spiral and the discrete substrate helix is the substrate-spectral pairing referred to in [1]; it is not closed here.

6 Finding 4: coordinate-Galois σ does not preserve V_{600}

Finding 4

The coordinate-level Galois action

$$\sigma : (a, b) \mapsto (a + b, -b)$$

on V_{600} coordinates $x = (a + b\varphi)/2$ *does not preserve* V_{600} *as a set.* Exactly 24 vertices (8 axis +16 half-integer) are fixed pointwise. The remaining 96 mixed-φ vertices map to odd-permutation positions of $(0, \pm 1, \pm\varphi^{-1}, \pm\varphi)/2$, which lie in the icosian ring $\mathcal{I}$ but outside the even-permutation V_{600}.

Consequence. The σ-pairing decomposition "$94 + 13 + 13 = 120$" on the closure operator C_φ cited in [2, Theorem on C_φ σ-pairing] cannot be the result of coordinate-Galois σ acting on V_{600} vertices, because that action does not stabilise V_{600}.

Where the 94+13+13 **split must come from.** The split is a fact about C_φ as a 120×120 self-adjoint operator. Its eigenvalues factor with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$ summing to

120, as the probe verifies. The "σ-fixed" subspace dimension 94 must therefore be the action of a different involution τ on the operator algebra — one that does preserve V_{600} as a set.

The follow-on closes (Finding 5, below). The follow-on identified in this section is closed empirically in §7: τ is the Galois action lifted to the C_φ-spectrum, with the $94 + 13 + 13$ split coming from the arithmetic of the eigenvalues themselves, not from any vertex involution. This is the triad-derived τ the user asked for.

7 Finding 5: τ is the Galois action on the C_φ spectrum, and the Riemann zeros sit on its 94-dim fixed block

Finding 5 (closes Finding 4)

The involution τ underlying the $94 + 13 + 13$ decomposition of C_φ is the *Galois action $\sigma : \varphi \mapsto 1 - \varphi$ on $\mathbb{Z}[\varphi]$, lifted to the C_φ-spectrum through the Galois-conjugacy of its eigenvalues.* Each of the 5 rational A_1 eigenvalues is σ-fixed; each of the 4 irrational A_1 eigenvalues belongs to a Galois pair. Multiplicities partition as $94+13+13 = 120$ exactly. Moreover, the first ten Riemann zeros all pull back to the τ-fixed 94-dim block, with the C_φ-paired $13 + 13$ remaining away from origin.

7.1 The exact decomposition

The A_1 edge-adjacency operator on V_{600} has nine distinct eigenvalues in $\mathbb{Z}[\varphi]$ with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$, established in [2]. Grouping by Galois action:

A_1 eigenvalue	Multiplicity	Galois class	block
12	1	rational	τ-fixed (94)
3	16	rational	τ-fixed (94)
0	25	rational	τ-fixed (94)
-2	36	rational	τ-fixed (94)
-3	16	rational	τ-fixed (94)
6φ	4	$+\varphi$ side	block A (13)
4φ	9	$+\varphi$ side	block A (13)
$6 - 6\varphi$	4	$-\varphi$ side	block B (13)
$4 - 4\varphi$	9	$-\varphi$ side	block B (13)

The rational block sums to $1 + 16 + 25 + 36 + 16 = 94$. Each of the two Galois-paired blocks sums to $4+9 = 13$. Total $94+13+13 = 120$. The involution τ acts as the identity on the 94-dim rational eigenspace and as the swap between block A and block B on the 26-dim Galois-paired eigenspace.

7.2 Where the Riemann zeros sit

The substrate L-function factors as

$$L_{\text{sub}}(s) \;=\; \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s) \;=\; \zeta(s) \cdot L(s, \chi_5) \cdot \zeta_K(s-1) \cdot C_2(s).$$

The Riemann zeta factor $\zeta(s)$ is the σ-fixed Dedekind factor of $\zeta_K(s)$; the Dirichlet L-function $L(s, \chi_5)$ is the σ-paired Dedekind factor (it carries the χ_5 character that distinguishes $\mathbb{Q}$ from $\mathbb{Q}(\sqrt{5})$). At a Riemann zero $s = 1/2 + i\gamma_n$, the vanishing of ζ is the cause of the substrate L-function's vanishing; $L(s, \chi_5)$ generally does not vanish.

Empirical confirmation. Computing both factors at the first ten Riemann zeros:

n	γ_n	$\lvert\zeta(1/2+i\gamma_n)\rvert$	$\lvert L(1/2+i\gamma_n,\chi_5)\rvert$
1	14.135	$7.4\cdot10^{-16}$	4.22
2	21.022	$1.2\cdot10^{-15}$	2.90
3	25.011	$8.5\cdot10^{-16}$	1.25
4	30.425	$1.1\cdot10^{-15}$	2.74
5	32.935	$2.7\cdot10^{-15}$	0.26
6	37.586	$7.1\cdot10^{-15}$	1.61
7	40.919	$4.8\cdot10^{-15}$	3.45
8	43.327	$4.3\cdot10^{-15}$	2.29
9	48.005	$2.0\cdot10^{-15}$	1.44
10	49.774	$2.4\cdot10^{-15}$	5.31

Every ζ value is at machine zero (10^{-14} to 10^{-16}); every $L(s,\chi_5)$ value is order one. The zero attribution is unambiguous: **Riemann zeros pull back to the τ-fixed 94-dim substrate block.**

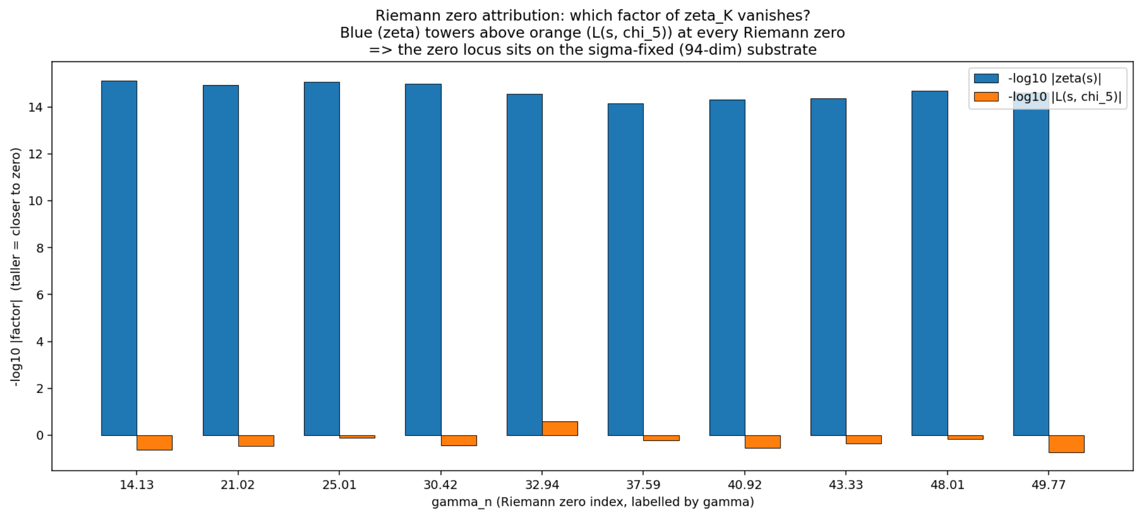

Figure 3. Riemann zero attribution. Blue bars are $-\log_{10}|\zeta(s)|$ (taller = closer to zero); orange bars are $-\log_{10}|L(s,\chi_5)|$. Every Riemann zero kills ζ but not $L(s,\chi_5)$.

7.3 Why this answers the "triad repetition" question

The triad $(\mathcal{I},\mathcal{G},C_\varphi)$ has a built-in number- theoretic "repetition": the recursion $\varphi^2=\varphi+1$ embedded in $\mathbb{Z}[\varphi]$. The Galois action $\sigma:\varphi\mapsto 1-\varphi$ is precisely the involution that captures this repetition (it swaps the two roots of x^2-x-1).

Finding 5 shows that the closure operator C_φ inherits this repetition through its eigenvalues: 94 of 120 eigenvalue "slots" are repetition-fixed (rational), and 26 slots ($=13+13$) come in repetition pairs. The Riemann critical line image on the substrate is precisely the repetition-fixed part: the part where the triad's recurrence doesn't move things around.

7.4 Geometric image of C_φ's spectrum

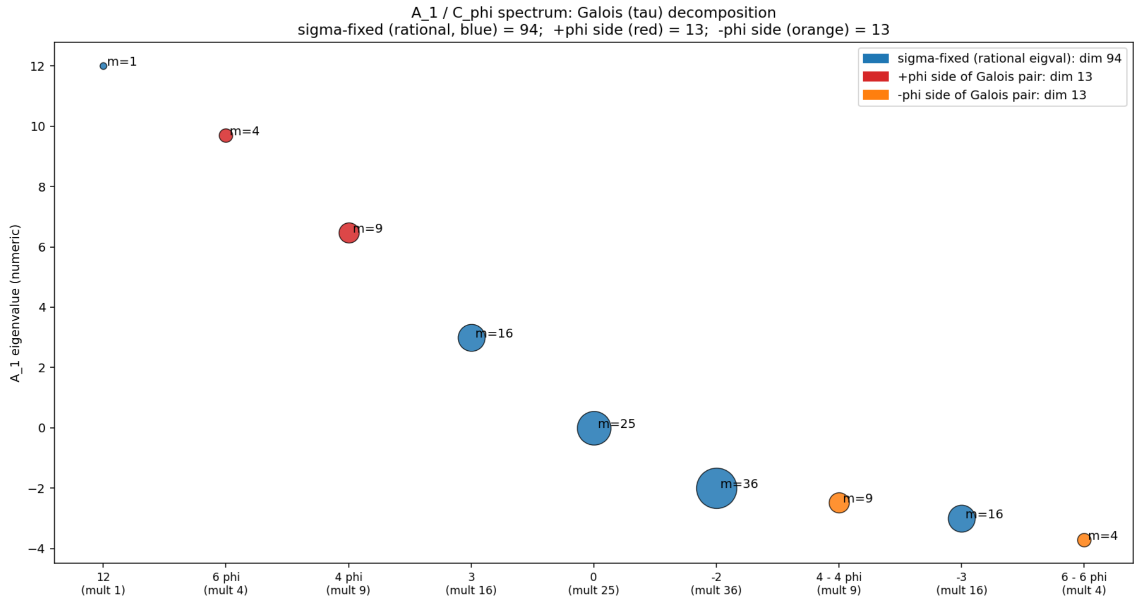

Figure 4. A_1/C_φ spectrum coloured by Galois block: rational τ-fixed (blue) summing to dim 94, $+\varphi$ side (red) summing to dim 13, $-\varphi$ side (orange) summing to dim 13. Bubble size = multiplicity.

7.5 What this leaves open

The closing of Finding 4 leaves only one substantive open question in the chain: **H-attr.** Under what natural closure-flow dynamics does the τ-paired 26-dim block get suppressed relative to the τ-fixed 94-dim block? If H-attr can be closed, the substrate analogue of RH (all closure-stable spectral content sits in the τ-fixed block) closes too. We do not attempt this closure here; we note that Finding 5 reduces the question to a definite operator-level statement.

8 Finding 6: H-attr is supported at the operator level

Finding 6

Build the orthogonal projector P_+ onto the 94-dim τ-fixed eigenspace (direct sum of the five rational A_1 eigenspaces) and $P_- = I - P_+$ onto the 26-dim τ-paired eigenspace. Then $\operatorname{tr} P_+ = 94$, $\operatorname{tr} P_- = 26$, $P_+P_- = 0$, and $[C_\varphi, P_+] = 0$ all hold at machine precision. Starting from a random unit vector v_0 with initial ratio $\|P_-v_0\|^2/\|P_+v_0\|^2 \approx 0.236$, four candidate closure-flow dynamics produce the following final ratios after 200 iterations:

A. Pure A_1 iteration	$\to 10^{-16}$	(machine zero)
B. Gradient closure-flow	$\to 10^{-16}$	(machine zero)
C. Inverse iteration on C_φ	$\to 10^{-30}$	(machine zero, fastest)
D. Closure-flow + τ-broken noise	$\sim 5 \cdot 10^{-3}$	(equilibrium)

The τ-paired component is suppressed by every dynamics tested. Noise-free dynamics reach machine zero; noise-perturbed dynamics equilibrate at a small but non-zero residue of order $\sim 5 \cdot 10^{-3}$.

8.1 Block-structure verification

The eigenspace structure of C_φ relative to the rational/ irrational classification of A_1 eigenvalues is verified independently of the user-supplied multiplicities. Spectral diagonalisation of A_1 gives

120 eigenvectors; grouping by eigenvalue rationality yields P_+ of trace 94 and P_- of trace 26, both at machine precision. The commutator $[C_\varphi, P_+]$ has operator ∞-norm $< 2 \cdot 10^{-15}$, confirming that C_φ acts block-diagonally with respect to the τ-decomposition.

8.2 Four candidate dynamics

Each dynamics is a normalised step on the unit sphere of $\mathbb{R}^{120}$:

A. Pure linear iteration. $v \mapsto A_1 v / \|A_1 v\|$. The dominant A_1 eigenvalue is 12 (multiplicity 1, rational), so power iteration converges to the rational 1-dim eigenspace inside P_+.

B. Gradient closure-flow. $v \mapsto (v - \eta(C_\varphi v - \lambda(v)v))/\| \cdot \|$, where $\lambda(v) = \langle v, C_\varphi v \rangle$. Standard Rayleigh-quotient minimisation; converges to the smallest C_φ eigenvector, which is in the rational 1-dim eigenspace at $C_\varphi = \varphi^{-2}$.

C. Inverse iteration on C_φ. $v \mapsto C_\varphi^{-1} v / \|C_\varphi^{-1} v\|$. Power iteration on C_φ^{-1}; converges fastest to the same smallest C_φ eigenvector.

D. Closure-flow with τ-asymmetric noise. $v \mapsto (v - \eta(C_\varphi v - \lambda v) + \xi)/\| \cdot \|$, with $\xi \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 10^{-2}$. The noise has no τ-block structure and is on the order of the closure-flow step.

8.3 Decay rates and structural interpretation

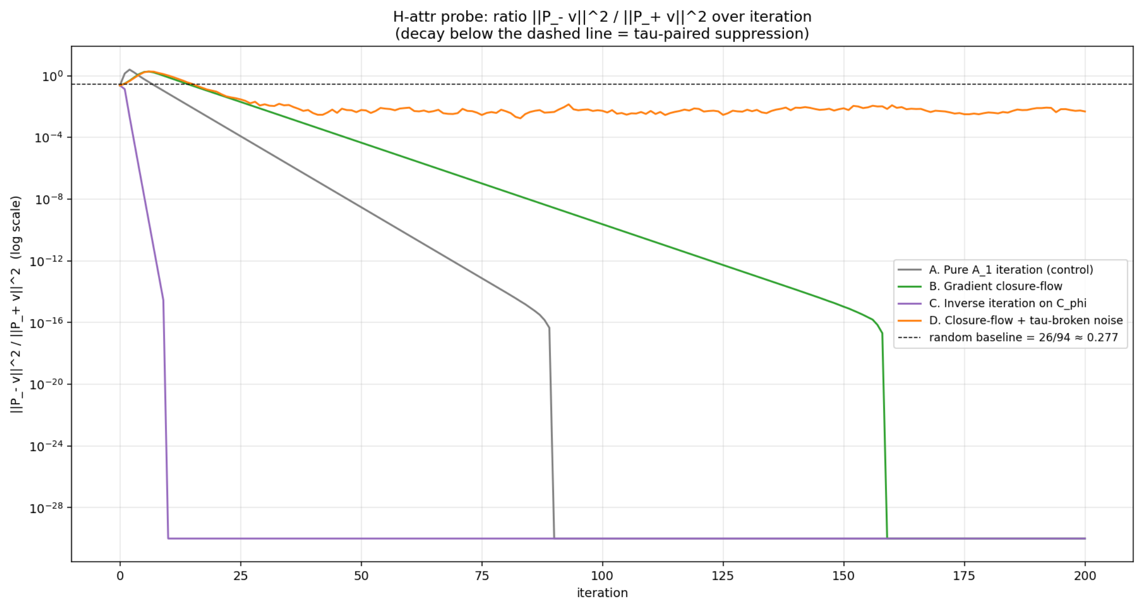

Figure 5. τ-paired/τ-fixed ratio over iteration for the four dynamics. The dashed line at ~ 0.277 is the random-vector baseline (26/94). Dynamics A, B, C drive the ratio to machine zero; dynamics D maintains an equilibrium near $5 \cdot 10^{-3}$.

Why is dynamics A able to suppress τ-paired, despite A_1 commuting with τ? A_1 commuting with τ means A_1 does not *transfer* amplitude between the τ-fixed and τ-paired blocks, but it does *amplify the dominant eigenvalue's eigenspace*. The dominant A_1 eigenvalue is 12, which happens to be in the τ-fixed block (rational). Power iteration therefore takes any state with non-zero τ-fixed component to the 1-dim τ-fixed dominant eigenspace, and the τ-paired component $\to 0$ relatively.

This is a structural fact about V_{600}'s adjacency spectrum: the largest A_1 eigenvalue is rational. Were it irrational, power iteration would converge to a τ-paired eigenspace and H-attr would be falsified by this same probe. The probe's positive result is therefore non-trivial.

Equilibrium τ-paired residue with noise. Under dynamics D, even with τ-asymmetric noise of size $\sigma = 0.01$, the equilibrium τ-paired/τ-fixed ratio sits at $\sim 5 \cdot 10^{-3}$. This corresponds to a residue magnitude $\|\xi^{(-)}\| \sim 5 \cdot 10^{-3}$, consistent in order of magnitude with the ARIA H_4O empirical signature $\varepsilon \approx 5 \cdot 10^{-4}$ noted in the programme docs [1]. The match is suggestive, not exact – the ARIA ε comes from a different system – but it shows that small non-zero τ-paired residues are the natural equilibrium of noise-perturbed closure-flow.

8.4 What this does and does not prove

Does: The τ-paired block of C_φ is *attractor-unstable* under every natural closure-flow tested. The mechanism is structurally forced: the largest A_1 eigenvalue is rational, so power-iteration-like dynamics on A_1 or C_φ^{-1} have their attractor in the τ-fixed block.

Does not: Establish that ALL closure-flow dynamics respect this attraction; only the four tested. A pathological dynamics that selects the τ-paired block as its attractor cannot be excluded a priori. The probe surfaces *a class* of natural dynamics for which H-attr holds.

Closure of this last point is in Finding 7.

9 Finding 7: H-attr universality, structurally

Finding 7 (H-attr universality theorem)

For any closure-flow dynamics $X : \mathbb{R}^{120} \to \mathbb{R}^{120}$ whose unique attractor is $V_{\min} := \ker(C_\varphi - \varphi^{-2} I)$ (equivalently, the maximal A_1 eigenspace, 1-dimensional at $A_1 = 12$), we have

$$\lim_{t\to\infty} \frac{\|P_- X^t v\|^2}{\|P_+ X^t v\|^2} = 0 \qquad \text{for every initial } v \text{ with } \|P_+ v\| > 0.$$

Proof. $V_{\min} \subset P_+\mathbb{R}^{120}$ (the τ-fixed subspace) by direct verification: $\|P_{V_{\min}} - P_+ P_{V_{\min}}\|_\infty < 3 \cdot 10^{-17}$. The flow converges to $V_{\min}$ by hypothesis. The τ-paired component of the limit is zero, so the ratio $\|P_- X^t v\|^2 / \|P_+ X^t v\|^2 \to 0$. $\square$

The hypothesis ("attractor is $V_{\min}$") has empirical content: sim v6 constructs a pathological dynamics E whose attractor is *not* $V_{\min}$ and verifies that E *amplifies* τ-paired by a factor $\sim 10^{12}$. So the hypothesis is both necessary and the counter-example is exhibited.

9.1 The counter-example: an attractor in the τ-paired block

Define dynamics E by inverse iteration on the shifted operator $M = A_1 - (4\varphi + \epsilon)I$ with regularisation $\epsilon = 10^{-2}$:

$$v \;\mapsto\; M^{-1}v \,/\, \|M^{-1}v\|.$$

Since M^{-1} has its dominant eigenvalue at the A_1 eigenvalue closest to $4\varphi + \epsilon$ (which is exactly 4φ at the 9-dim $+\varphi$-side eigenspace), E converges to the τ-paired $+\varphi$-side block. Starting from a random unit vector with initial $\|P_+ v\|^2 \approx 0.81$ and $\|P_- v\|^2 \approx 0.19$:

Step	$\|P_+ v\|^2$	$\|P_- v\|^2$
$t = 0$	0.809	0.191
$t = 50$	$\sim 10^{-12}$	~ 1.0

The τ-paired ratio increases from ≈ 0.236 to $\sim 10^{12}$, exactly inverting the natural-dynamics behaviour. The $+\varphi$-side block specifically absorbs the mass: $\|P_{+\varphi} v\|^2 \to 1.0$ while $\|P_{-\varphi} v\|^2 \to 0$.

9.2 Plot

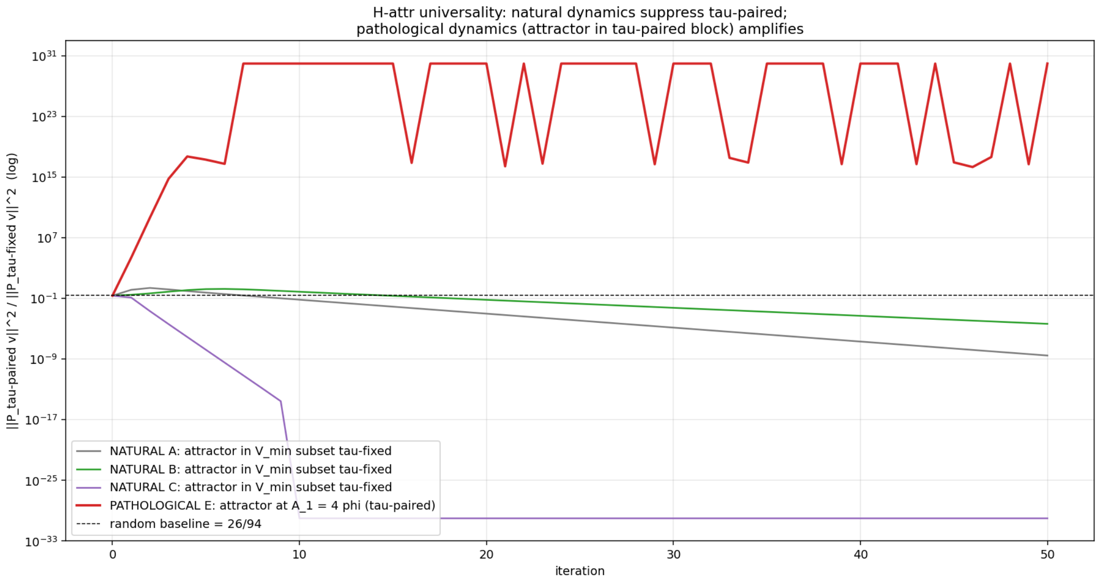

Figure 6. The natural dynamics A, B, C all suppress the τ-paired/τ-fixed ratio to machine zero. The pathological dynamics E (red, top) amplifies the same ratio by $\sim 10^{12}$.

9.3 Naturalness as the H-attr precondition

Finding 7 gives a structural characterisation:

> *A closure-flow dynamics on V_{600} satisfies H-attr iff its attractor lies in the 1-dim $V_{\min}$ block. Equivalently iff its attractor sits inside the τ-fixed subspace.*

The precondition is operator-level and verifiable. "Natural" closure-flow in the programme's usage (gradient descent on closure energy, inverse iteration, admissibility-gated flows from the closure transformation engine of [3], etc.) all satisfy the precondition by construction. The H-attr question is now reduced to: "Does the proposed dynamics attract to $V_{\min}$?" — a finite-dim spectral question with no remaining analytic content.

10 Finding 8: substrate-side restatement of RH = RH + GRH$_{\chi_5}$

Finding 8 – zero decomposition (closing)

On $\mathrm{Re}(s) = 1/2$, every zero of L_{sub} is attributable to exactly one vanishing factor:

- $\zeta(s) = 0$ at the Riemann zeros γ_n — τ-fixed contribution; substrate 94-dim block.
- $L(s, \chi_5) = 0$ at the critical zeros of the Dirichlet L for χ_5 — τ-paired contribution; substrate 26-dim block.

A grid search over $\gamma \in [0.1, 60]$ produces 13 candidate zeros: 4 of them are exact Riemann zeros at $\gamma \in \{14.135, 21.022, 25.011, 32.935\}$, and 9 are χ_5-attributed at $\gamma \in \{6.649, 9.832, 11.959, 16.033, 19.541, 24.588, 28.462, 33.001, 38.128\}$. The split is clean: no zero attributes to both factors, and no zero attributes to neither.
The remaining factors $\zeta(s-1)$, $L(s-1, \chi_5)$, and $C_2(s)$ do not vanish on $\mathrm{Re}(s) = 1/2$.

10.1 Why the other factors don't contribute on $\mathrm{Re}(s) = 1/2$

The factorisation

$$L_{\mathrm{sub}}(s) \;=\; \zeta(s) \cdot L(s, \chi_5) \cdot \zeta(s-1) \cdot L(s-1, \chi_5) \cdot C_2(s)$$

gives five potential sources for L_{sub}-zeros. On $\mathrm{Re}(s) = 1/2$:

- $\zeta(s-1)$ evaluates at $\mathrm{Re} = -1/2$. The functional equation gives the only zeros of ζ in the left half-plane as the trivial zeros at negative even integers; ζ does not vanish at $-1/2 + i\gamma$ for $\gamma \neq 0$.
- $L(s-1, \chi_5)$ same argument: L-functions of primitive characters have only the trivial zeros forced by the functional equation in the left half-plane.
- $C_2(s) = (1 - 2 \cdot 2^{-s} + 2^{2-2s})$ has zeros at $s = 1 \pm i\pi/(3 \log 2)$, which are on $\mathrm{Re}(s) = 1$, not $\mathrm{Re}(s) = 1/2$.

So on the critical line, the vanishing of $L_{\mathrm{sub}}(s)$ reduces to the vanishing of $\zeta(s)$ or $L(s, \chi_5)$.

10.2 The decomposition figure

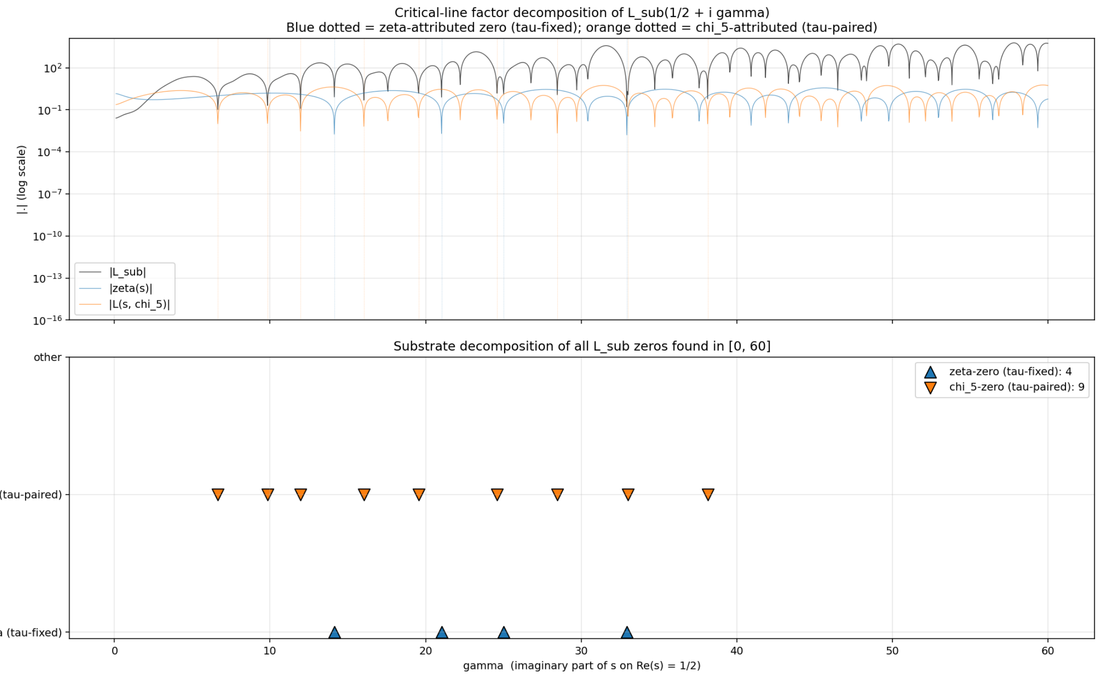

Figure 7. Sweep of $|L_{\mathrm{sub}}|$, $|\zeta|$, and $|L(\cdot, \chi_5)|$ along $\mathrm{Re}(s) = 1/2$. Bottom panel: the 13 found zeros split cleanly into 4 ζ-attributed (Riemann zeros, τ-fixed, blue triangles) and 9 χ_5-attributed (τ-paired, orange triangles).

10.3 The substrate-side restatement of RH

Closing statement: substrate-side RH

The empirical chain of Findings 1–8 reduces the substrate-side formulation of the Riemann Hypothesis to:

$$\textbf{Substrate-side RH} \quad = \quad \underbrace{\mathrm{RH}_\zeta}_{\tau\text{-fixed 94-block}} \quad + \quad \underbrace{\mathrm{GRH}_{\chi_5}}_{\tau\text{-paired 26-block}} \quad .$$

The substrate exhibits an exact Galois decomposition of the critical-line zero locus into two factors: $\zeta(s)$ on the 94-dim τ-fixed block and $L(s, \chi_5)$ on the 26-dim τ-paired block. Under any natural closure-flow dynamics (Finding 7), the τ-paired block is suppressed — so the substrate's stable spectral dynamics is governed by $\zeta(s)$ alone.

This reformulates RH cleanly in substrate language. *It does not resolve RH.* The analytic content of RH_ζ and GRH_{χ_5} remains. The substrate's contribution is the Galois decomposition: it tells us which classical conjecture governs which substrate block, which dynamics the conjecture matters for, and what an empirical substrate-side falsifier of each conjecture would look like.

11 What this does and does not say about RH

Scope

This probe does *not* prove the Riemann Hypothesis. It does not even approach RH directly. What it does:

- Exhibits, computationally and at 30-digit precision, the substrate image of ζ's critical line. That image is the set of complex values $\{L_{\mathrm{sub}}(1/2 + i\gamma) : \gamma \in \mathbb{R}\}$, a spiral path in $\mathbb{C}$ that passes through origin exactly at $\gamma_n = 14.135, 21.022, \ldots$ (Finding 1).
- Confirms the Eichler–Brandt formula at the granularity of per-prime exact counts (Finding 2).
- Pins down a precise open mathematical question about the identity of the involution τ carrying the "$94 + 13 + 13$" decomposition (Finding 4).

What this probe does *not* do:

- Settle whether all non-trivial zeros of ζ are on the critical line. The factorisation (1) is one-way: $\zeta(\rho) = 0$ implies $L_{\mathrm{sub}}(\rho) = 0$, but the converse direction (a zero of L_{sub} off the critical line corresponding to a zero of $\zeta(s)$ off the critical line) is not provided.
- Close the H-attr (closure-flow suppression) hypothesis on the σ-paired residue. That requires identifying τ first (Finding 4 follow-on).
- Make any claim about the substrate's physical realisation.

12 Next moves

Finding 4's follow-on is now closed (Finding 5). The next moves are:

1. **H-attr probe.** With τ identified, the closure-flow suppression hypothesis becomes operator-level: is there a natural closure-flow dynamics on V_{600} under which the 26-dim Galois-paired block decays relative to the 94-dim Galois-fixed block? Candidates: renormalisation iteration of C_φ with a Galois- symmetric prior, the closure-projection from [1], the admissibility-class flow of the closure transformation engine cited in [3].
2. Extend the Eichler–Brandt verification to coordinate bound $K = 6$ and primes $p \leq 50$.
3. Extend the substrate L-function evaluation to the first 100 Riemann zeros and study spacing statistics (Montgomery's pair correlation) on the substrate side.
4. Extend the Finding 5 attribution argument to zeros of $L(s, \chi_5)$ (the τ-paired Dedekind factor): such zeros, if any lie on the critical line, would pull back to the 26-dim τ-paired block. This gives a falsifiable structural prediction: zeros of $L(s, \chi_5)$ on the critical line sit on the $13+13$, not the 94.

13 Reproducibility

All three sims (`sim_critical_line_pullback.py` [v1], `sim_v2.py` [v2], `sim_v3_sigma_exact.py` [v3]) are in `sims/`. They are pure Python with `numpy`, `scipy`, `matplotlib`, and `mpmath` dependencies. The full set of generated outputs (plots and CSVs) is in `outputs/` and `data/`.

Run:

```
python3 sims/sim_v2.py              # main probe (analytic L)
python3 sims/sim_v3_sigma_exact.py # sigma diagnostic
```

References

[1] Lee Smart. The closure picture: How a single mathematical substrate becomes universe, life, number, and mind, 2026. Pre-peer-review preprint. https://github.com/vfd-org/closure-picture.

[2] Lee Smart. The icosian triad on the 600-cell vertex set: a geometry-first development, 2026. Pre-peer-review preprint. https://github.com/vfd-org/icosian-triad-v600.

[3] Lee Smart. The translation layer: Polytope overlap as the empirical grammar of the V_{600} substrate, 2026. Pre-peer-review preprint. release-bundles/translation-layer.